

MOTOR VEHICLE  
COLLISIONS(VISION ZERO)

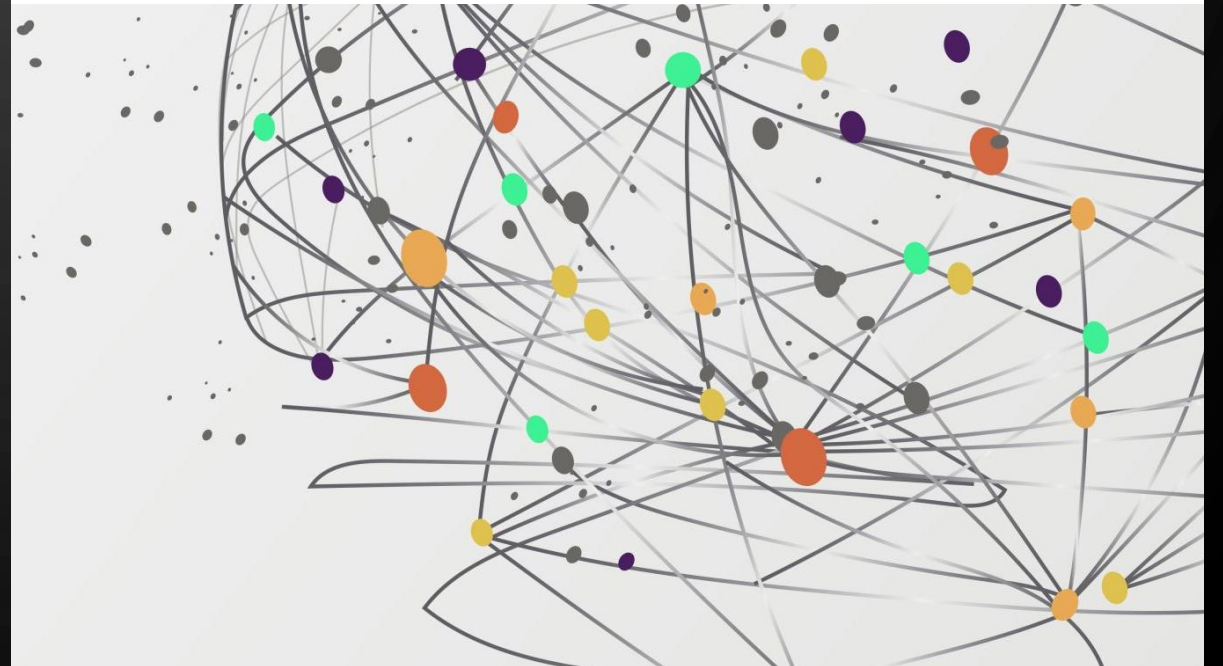
CAPSTONE PROJECT  
LAGUARDIA COMMUNITY  
COLLEGE  
DC 37 DATA ANALYTICS  
COHORT 1

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# VISION ZERO

## Building a Safer City



# RESEARCH QUESTIONS

- Which Borough has the highest Collisions?
- Which vehicle type occurred more frequently?
- The impact of Vision Zero on the average number of collisions per year in Brooklyn

# BACKGROUND

- In 2025 there was over 47,000 vehicle related injuries
- Almost 7% lower than 2024 according to Queens Chronical
- Thanks partially due to the Vision Zero incentive
- To analyze motor vehicle collisions

# DATA SOURCE

## NYC OPEN DATA/MOTOR VEHICLE COLLISIONS-CRASHES

| About this Dataset |                       |                          |  |
|--------------------|-----------------------|--------------------------|--|
| Updated            |                       | Data Collection          |  |
| May 6, 2026        |                       | Motor Vehicle Collisions |  |
| Data Last Updated  | Metadata Last Updated | Update                   |  |
| May 6, 2026        | April 19, 2021        | Update Frequency         |  |
| Date Created       |                       | Automation               |  |
| April 28, 2014     |                       | Date Made Public         |  |
| Views              |                       | Dataset Information      |  |
| 855K               |                       | Agency                   |  |
| Downloads          |                       | Police Department (NYPD) |  |
| 370K               |                       |                          |  |

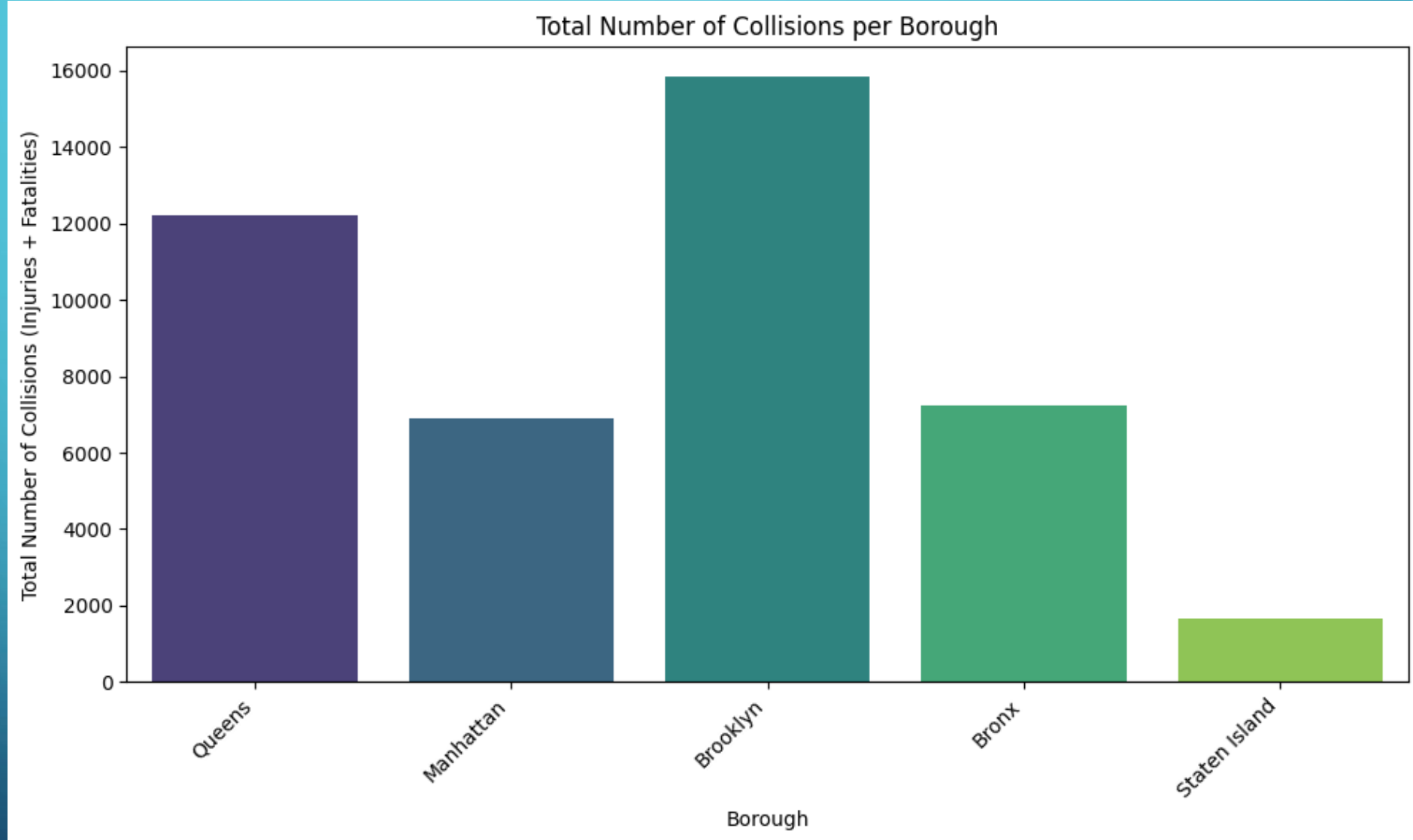
# METHODOLOGY

- Downloaded 100k rows of the data through colab
- Reviewed the data in detail to determine the rows and columns to keep for my analysis
- Cleaned data to exclude missing/null values, made subsets using proper syntaxes
- Calculated the total number of collisions that occurred in each borough
- Calculated most frequent vehicle types involved in collisions across all boroughs
- Created Bar charts to visualize the analysis

# FINDINGS

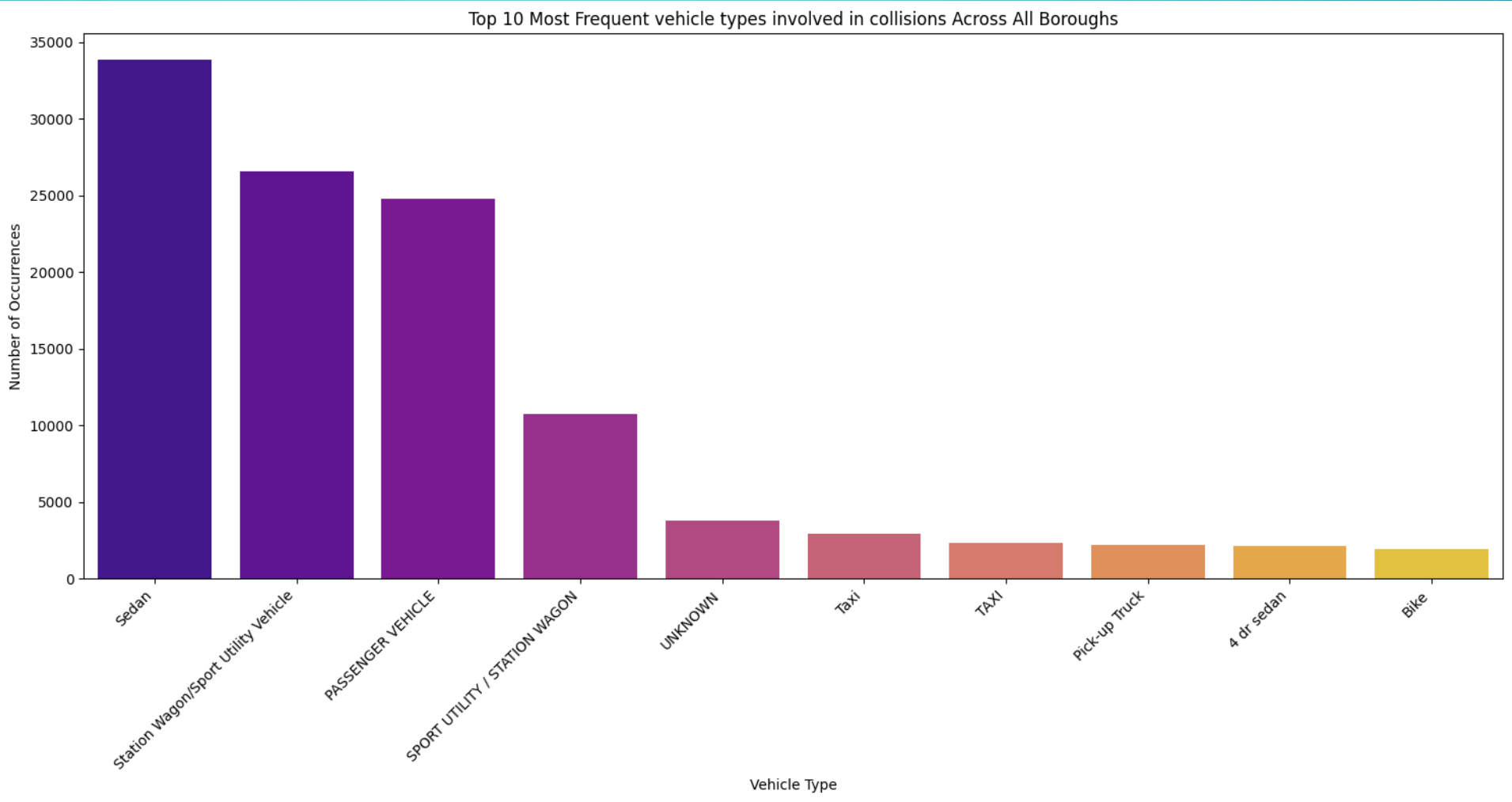
Brooklyn  
has most  
number of  
collisions

Staten  
Island has  
least  
number of  
collisions



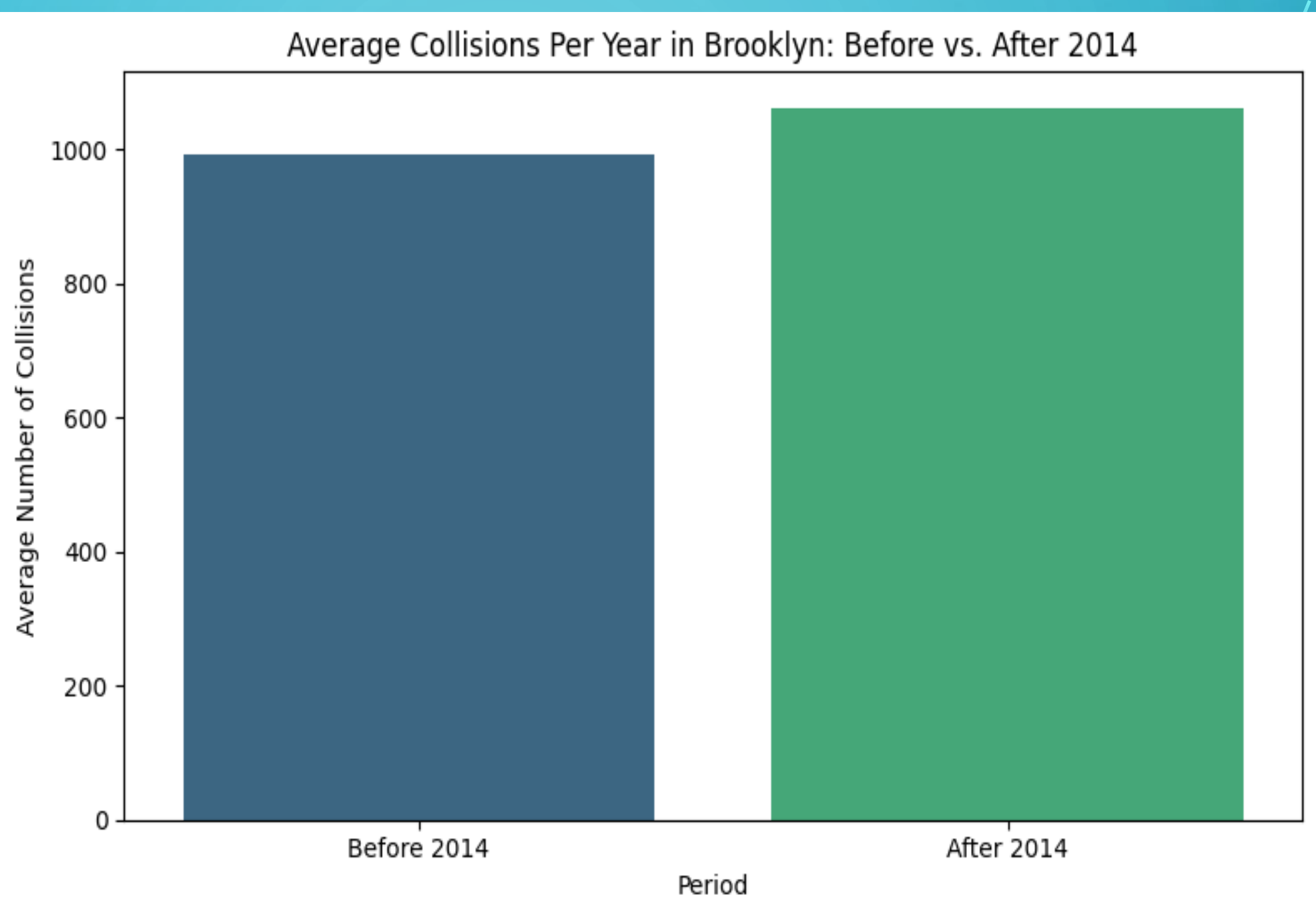
# FINDINGS

Across all the boroughs, most collisions involved sedans.



# FINDINGS

- Less datapoints before 2014
- Graph doesn't match expected conclusion and actual data from news articles





# CONCLUSIONS AND FUTURE OUTLOOK

- Most collisions occurred in Brooklyn between 2012 and 2026
- Sedans were involved in majority of collisions
- Correlation between number of collisions and Vision Zero initiative

# REFERENCES

- [Motor Vehicle Collisions - Crashes | NYC Open Data](#)
- <https://www.nyc.gov/html/dot/html/pr2025/decline-in-traffic-deaths.shtml>